

Chapter 9: Measurement Bias & “Noncausal” Diagrams

Causal Inference in Practice: Retention Offer Use Case (“Did the Retention Offer Actually Reduce Churn?”)

1. Measurement Bias: When Measured Variables Differ from True Variables

In observational studies and business analytics, we rarely observe true underlying causal constructs directly. Instead, we record proxy measurements or recorded data in CRM databases (*Section 9.1, pp. 119–121*).

- **True Causal Variables vs. Measured Proxies:** Let A be true treatment assigned, Y be true outcome, and L be true baseline confounders. We observe measured versions A^* , Y^* , and L^* .
- **Definition of Measurement Bias:** Measurement bias occurs when the association between A^* and Y^* differs from the true causal effect of A on Y due to measurement error in A , Y , or L .
- **Retention Offer Context:**
 - True A : Customer actually received and read the retention discount email.
 - Measured A^* : CRM log indicating email was sent (some emails landed in spam).
 - True Y : Customer genuinely abandoned service/brand.
 - Measured Y^* : Account flag marked “inactive” in month 3 (misclassifying paused or ported accounts).

2. Non-differential vs. Differential Measurement Error

Measurement error is classified based on whether the error in one variable depends on the values of other variables in the causal mechanism (*Sections 9.2 & 9.3, pp. 121–125*).

Taxonomy of Measurement Error

Non-differential Error (Section 9.2): Measurement error in A^* is independent of true outcome Y (given true A), or error in Y^* is independent of true treatment A (given true Y).

- *Impact:* Classically leads to **attenuation** (bias toward the null), underestimating the true retention effect.
- *Exception:* In multi-category or non-binary settings, non-differential error can sometimes bias away from the null.

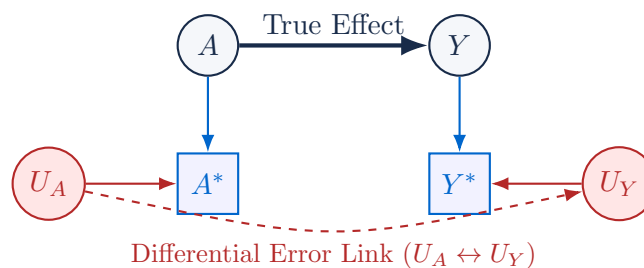
Differential Error (Section 9.3): Measurement error in A^* depends on outcome Y , or error in Y^* depends on treatment A .

- *Example:* Customers who churned ($Y = 1$) actively recall receiving a promotional offer ($A^* = 1$) differently than retained customers ($Y = 0$) (recall bias).
- *Impact:* Can cause severe bias in **any direction** (overestimating, underestimating, or reversing the sign of the effect).

3. Measurement Bias in Causal Diagrams (DAGs)

Causal DAGs represent measurement error by explicitly adding separate nodes for true variables (A, Y, L), measured variables (A^*, Y^*, L^*), and measurement error terms (U_A, U_Y, U_L) (Section 9.4, pp. 125–128).

Figure 9.1: Causal DAG with Treatment & Outcome Measurement Error



- **Non-differential:** U_A and U_Y are independent (no red dashed link).
- **Differential:** U_A and U_Y share a common cause or direct link, creating a non-causal association path $A^* \leftarrow U_A \rightarrow U_Y \rightarrow Y^*$.

4. Misclassification of Treatment & Intention-to-Treat (ITT)

In randomized experiments or marketing campaigns, measurement error in treatment implementation leads directly to the concept of **Intention-to-Treat (ITT)** vs. **Per-Protocol** effects (Section 9.5, pp. 128–130).

ITT Effect vs. True Treatment Effect

- **Assigned Treatment (Z):** Random offer email dispatched by CRM.
- **Received Treatment (A):** Customer opens, reads, and applies the promo code.
- **Intention-to-Treat (ITT) Effect:** Measures the causal effect of *offering* the discount ($Z \rightarrow Y$).
 - Unbiased estimate of policy/campaign effectiveness.
 - Consistently conservative (attenuated) estimate of the efficacy of actually applying the discount.
- **Per-Protocol Effect:** Measures the true causal effect of *receiving/using* the discount ($A \rightarrow Y$). Requires adjustment for non-compliance confounding ($Z \rightarrow A \leftarrow U \rightarrow Y$).

5. Summary Actionable Recommendations for Data Teams

Bias Type	Causal Mechanism in CRM Data	Correction / Mitigation Strategy
Confounding	Common causes of offer assignment and churn ($A \leftarrow L \rightarrow Y$).	IP Weighting, Standardization, Propensity Scores.
Selection Bias	Conditioning on common effects / colliders ($A \rightarrow [C] \leftarrow Y$).	IP Weighting for Censoring (W^C), avoid post-treatment conditioning.
Measurement Error	Discrepancy between logged logs (A^*, Y^*) and true reality (A, Y).	Validation subsamples, Quantitative Bias Analysis, ITT framework.